



DRIVING THE EXASCALE TRANSITION



CERTIFICATE OF ATTENDANCE

This acknowledges that

Gonzalo Nicanor Molina

Attended the course

Computational Spectroscopy: Hands-on Many-Body Calculations with the Yambo Code

held on 5–7 May 2026

Lucie

Roznovska

Digitálně podepsal
Lucie Roznovska
Datum: 2026.05.22
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Lucie Rožnovská

Training Representative of the Czech National Competence Centre in HPC



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the European Union



Ministry of Education,
Youth and Sports
of the Czech Republic



EuroHPC
Joint Undertaking

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Certificate of attendance in the From Building to City Energy Simulation course

Successful completion of the 10,5-hour course Computational Spectroscopy: Hands-on Many-Body Calculations with the Yambo Code

Course content level

Beginner, intermediate

Topics covered

- Introduction: MaX Ecosystem, Yambo, and the Motivation for
- Theory: Linear Response Theory, Independent Particle (IP), and RPA
- Hands-on I: Yambo Environment Setup, Data Interface, and Initial Runs
- Theory: Foundations of the GW approximation
- Practice: GW in Production —Algorithms and Approximations
- HPC: Parallelism Strategy and GPU Porting in Yambo
- Hands-on II: Computing Quasiparticle Band Structures
- Theory: The Bethe-Salpeter Equation and BSE Solvers
- HPC Strategy: Maximizing Energy Efficiency at the
- Hands-on III: Optical Absorption and Exciton Analysis

Course format

Online lectures, demos, and hands-on labs (10,5 hours total, breaks excluded).

Lectures taught by

Daniele Varsano, Giacomo Sesti, Andrea Ferretti, Matteo D'Alessio, Claudia Cardoso, Nicola Spallanzani, Fulvio Paleari, Lubomír Říha

Organised by

[MaX CoE, NCC Czechia, NCC](#)

Course description

<https://events.it4i.cz/event/393/>